PhysioGuide Project Kickoff & 2-Week Sprint Plan

Hi guyz. The goal of this document is to align our efforts from day one. It outlines our mission, clarifies each person's role and core responsibilities, and provides a detailed, task-by-task breakdown for our first two-week sprint. Let's get started.

Our Mission: The Minimum Viable Product (MVP)

Our primary objective is to develop a functional Minimum Viable Product (MVP) by the December deadline outlined in our proposal. For the MVP, we will narrow our focus to ensure we deliver a polished and functional core experience.

MVP Scope:

- **Core Functionality:** A basic but functional VR gym environment where a user can perform exercises and receive real-time feedback.
- **Exercise Focus:** We will concentrate on perfecting **two** key upper-body exercises (e.g., Shoulder Abduction and Elbow Flexion) or maybe neck.
- **Technology Stack:** The system will integrate OpenPose for motion tracking with a live webcam feed, a Python backend for data processing, and a Unity VR application for the user experience on the Oculus Quest 2.
- **Feedback System:** The MVP will provide immediate feedback to the user through on-screen color changes (green for correct form, red for incorrect) and simple audio cues.

Team Roles & Responsibilities

To ensure clarity and ownership, here is the breakdown of our roles. While we will all collaborate closely, each member is the Directly Responsible Individual (DRI) for their domain.

Member	Role	Core Responsibilities
Taha Khan	**Project Lead**	Overall project management, version control (Git), scheduling, and ensuring seamless integration between the Python and Unity components.
Fatima Mehdi	**R&D Specialist**	Researching the biomechanics of our two target exercises, defining the rules for "correct" vs. "incorrect" form, and providing the technical specifications for the feedback system.
Javeria Anwer	**Unity & VR Developer**	Setting up the Unity development environment for the Oculus Quest 2, building the virtual gym, implementing the user interface, and programming the VR interactions in C#.
Hussain Kazmi	**OpenPose Specialist**	Building and configuring the OpenPose library, writing the Python scripts to capture webcam video, and processing the video to extract skeletal keypoint data in real-time.

Action Plan: Weeks 1-2 Sprint

This initial two-week sprint is focused on **foundational setup and learning**. The goal is for each of us to get our development environments fully operational and complete a "Hello, World" style task. This will ensure we have the tools and basic knowledge required before we tackle more complex integration tasks.

Week 1: Environment Setup & Project Initialization

Objective: Install all required software and establish the project's infrastructure.

Taha's Tasks:

1. **Create GitHub Repository:** Set up a new private repository for our project. Create a main branch and initial feature branches (vr-environment, openpose-module).
2. **Invite Team:** Add all team members as collaborators to the repository.
3. **Schedule Weekly Sync:** Schedule a recurring 1-hour weekly meeting for us to sync up, discuss roadblocks, and plan the week ahead.

Ye 1 day main bhi hojyga,

Fatima's Tasks:

1. **Begin Literature Review:** Start the research process as outlined in our proposal.
2. **Initial Exercise Research:** Focus on our first two target exercises: **Shoulder Abduction** and **Elbow Flexion**. Gather information on the correct biomechanics, key joints involved, and common patient errors during rehabilitation.

Jo video bheji hai wo dekhna, sir ne bheji hai

Javeria's Tasks:

1. **Install Unity Hub & Editor:** Download the Unity Hub and install a stable Long-Term Support (LTS) version of the Unity Editor (e.g., 2022.3 LTS or newer).
2. **Include Android Build Support: Crucially**, during the Unity Editor installation, you must select and include the "Android Build Support" module. The Oculus Quest 2 runs on Android, and we cannot build for the headset without this.
3. **Install Meta Quest Developer Hub (ODH):** Download and install the ODH from the Meta developer website. This tool will help manage the headset and deploy our application builds.
4. **Enable Developer Mode:** Follow the official Meta documentation to enable Developer Mode on the Oculus Quest 2 headset using the mobile app. This is a mandatory step for development.

Hussain's Tasks (OpenPose Specialist):

1. **Install Prerequisites:** Begin the installation of all prerequisites for building OpenPose from source on Windows 10.
2. **Visual Studio 2019:** Install Visual Studio 2019, ensuring the "Desktop development with C++" workload is selected.

3. **CMake:** Download and install the CMake GUI application.
4. **NVIDIA Dependencies (If Applicable):** If your development machine has a compatible NVIDIA GPU, download and install the specific versions of the CUDA Toolkit and cuDNN library required by OpenPose. If not, we will proceed with a CPU-only build. *Note: This process can be complex, so starting early is key.*

Ye wala kam Very IMP hai.

Week 2: Foundational "Hello, World" Tasks

Objective: Validate our development environments by creating and running a minimal proof-of-concept.

Javeria's Tasks:

1. **Create "Hello, VR" Project:**
    - Create a new 3D Unity project.
    - Using the Package Manager, import the **Meta XR Core SDK** from the Unity Asset Store into your project.
    - Create a new scene and add the OVRCameraRig (or equivalent XR Origin) prefab. This object represents the VR headset and controllers in the virtual world.
2. **Build and Run on Quest 2:**
    - Connect the Oculus Quest 2 to your PC via a USB-C cable and enable Oculus Link.
    - In Unity's Build Settings, switch the platform to Android.
    - Build and run the empty scene.
    - **Success Criterion:** You can put on the headset and see your simple Unity scene. This confirms the entire toolchain is working correctly.

Hussain's Tasks:

1. **Configure OpenPose Build:**
    - Clone the OpenPose repository from GitHub.
    - Launch the CMake GUI. Point it to the OpenPose source code and a new build directory.
    - During configuration, find and enable the **BUILD_PYTHON** flag. This is essential for creating the Python API we need.
2. **Generate & Compile:**
    - Generate the Visual Studio solution files from CMake.
    - Open the generated .sln file in Visual Studio 2019.
    - Switch the build configuration from Debug to Release and compile the entire solution.

- **Success Criterion:** The project compiles successfully without errors. This is a major milestone.

Fatima's Tasks:

1. **Finalize Exercise Selection:** Confirm that Shoulder Abduction and Elbow Flexion are the best initial candidates for our MVP.
2. **Document Biomechanics:** For each exercise, begin creating a detailed specification document. This should include:
    - The primary anatomical joints to be tracked (e.g., shoulder, elbow, wrist).
    - The key angles that define correct posture (e.g., the angle between the upper arm and torso).
    - A description of the correct plane of movement.

Let's use our communication channels to stay in sync and help each other with any issues that arise. I'm looking forward to building this with all of you. Wese to, this is a 2 week plan but koshish ye hai k ye 1 week main ho kaam.